

Austin Jiang

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EXPERIENCE

DeepInfra

Software Engineer Intern | Python, PyTorch, Triton

Palo Alto, CA

Incoming Jan 2027 – Apr 2027

- Will work on inference acceleration for open-source large language models.

Noah's Ark Lab, Huawei Canada

Research Intern | Python, PyTorch

Toronto, ON

May 2026 – Present

- Designed an efficient 3D reconstruction model using residual-update gating for adaptive token early-halt, achieving a **57% latency reduction** and **11% depth accuracy improvement** on the 7-Scenes dataset.
- Developed a token aggregation framework to retain pruned tokens, limiting camera pose estimation **loss to < 1%**.
- Proposed a budget-controllable routing knob with dynamic loss reweighting, reducing overall training costs to **1/3**.

Multicore Lab, University of Waterloo

Undergraduate Research Assistant (Supervised by Dr. Trevor Brown) | C++, CUDA

Waterloo, ON

Sep 2025 – Present

- Developed a concurrent versioned pointer adapter within the benchmark, boosting throughput by **~2x**.
- Integrated a GPU B-Tree batching update mechanism, accelerating benchmark prefilling speed by **370%+**.
- Implemented high-throughput B+ Tree and ART Tree variants, supporting **10M-key** queries under **100+ threads**.

Wolfram Research

Research Intern (Part-time) | Mathematica

Remote

Sep 2025 – May 2026

- Engineered a multi-threaded halo-exchange framework for **800K+** keys across **8 cores**, cutting runtime by **56%**.
- Implemented sparse updates and dense scans via dynamic load-balancing, boosting throughput by **27%**.
- Authored a first-author paper on parallel algorithms, presented at the **Wolfram Technology Conference 2025**.

PROJECTS

Honk Pack, HackMIT 2026 | Python: Architected an end-to-end per-object 3D reconstruction pipeline using SAM3, Depth Anything 3, and ShapeR from phone video, coupled with a custom 3D bin-packing algorithm.

Rewrite, TreeHacks 2026 | Python: Built a fully interactive real-time 3D VR game leveraging Hunyuan WorldPlay 1.5, deploying tensor parallelism on Modal across 4x H200 GPUs to boost frame rates from 1 to 12 FPS.

ADaptiv, CMU NexHacks 2026 | Python: Built a video object replacement pipeline in ComfyUI using Gemini, SAM3, and Wan2.1, optimizing execution on an RTX 4090 to achieve a 3x speedup.

EDUCATION

University of Waterloo

Bachelor of Computer Science (Honours) | Major GPA: 4.0/4.0

Waterloo, ON

Expected Graduation: Apr 2028

- **Scholarships:** René Descartes National Scholarship (\$25,000), President's Scholarship of Distinction (\$2,500)
- **Coursework:** Data Structures, Algorithms, Development Tools, Compilers, Object-Oriented Programming

SKILLS

Languages: C++, Python, C, Unity (C#), Java, Bash, TypeScript

Tools/Libraries: PyTorch, Linux, Git, CUDA, CMake, NumPy, SQL, Docker, Nginx, Flask, FastAPI

Concepts: Transformers, Large Language Models, Concurrency, Compilers, Distributed Systems, Databases

ACHIEVEMENTS

Canadian Computing Olympiad 2026: Authored Day 1 Problem 3 (Contestant average score: 0.7/25).

Discover Citadel Program: Invited as 1 of ~100 students (< 1% acceptance rate).

Jane Street First-Year Trading and Technology Program: Invited as 1 of ~100 students (< 1% acceptance rate).

Generation Google Scholarship: Google's flagship global undergrad scholarship for impact in technology (1 of 55).

Canadian Computing Olympiad 2024 & 2025: Silver medalist x2 (5th and 6th out of 10,000+ participants).

USA Computing Olympiad 2024: Achieved the platinum division, ranked top 100 out of 15,564 participants.